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View Article Online
DOI: 10.1039/C6CP02409B

Journal Name

ARTICLE

New Boron Based Salts for Lithium-ion Batteries Using Conjugated Ligands

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

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A new anion design concept, based on combining a boron atom as the central atom and conjugated systems as ligands, is presented as a route for finding alternative Li-salts for lithium-ion batteries. The properties of a wide range of this way designed novel anions have been evaluated by DFT calculations focusing on three different fundamental success factors/measures: the strength of the cation-anion interaction, ultimately determining both the solubility and the ionic conductivity, the oxidation limit, determining the possible use *vs.* high voltage cathodes, and the reduction stability, revealing a possible role of the anion in the SEI-formation at the anode. For a few anions superior properties *vs.* today existing or suggested anions are predicted, especially the very low cation-anion interaction strengths are promising features. The design route itself shows to be versatile in determining the correlation between different choices of ligands and resulting overall properties – where the most striking feature is the decreased in lithium cation interaction energy upon using the (1Z,3Z)-buta-1,3-diene-1,2,3,4-tetracarbonitrile ligands. This also opens for further design of novel anions beyond those with a boron central atom.

1. Introduction

Ever since the commercialization of rechargeable lithium-ion batteries (LIBs), lithium hexafluorophosphate ($LiPF_6$) is the most commonly used electrolyte salt, owing to a well-balanced set of properties;[1] high salt dissociation degree and thereby high ionic conductivity of its carbonate solvent based electrolytes, wide electrochemical stability windows (ESWs) - stability towards oxidation and reduction or meta-stability leading to the formation of a protective layer, and compatibility towards other battery components such as solvents, active materials, and current collectors. However, LiPF6 also has severe safety problems that are difficult to mitigate completely and that are likely contributors to life-length limits of today's LiBs; it is very easily hydrolyzed, eventually resulting in the release of hazardous HF, and has poor thermal stability causing continuous electrolyte degradation.[2-6] Therefore, new lithium salts are urged for, but any such salt must outperform $LiPF_6$ with respect to many of the above requested properties to be interesting as a replacement.

Luckily, many of the above requested basic properties can be successfully estimated by standard computational chemistry methods, enabling us to search for new promising Li-salts/anions by in silico screening methods and thereby saving time and money by complementing/replacing costly exploratory synthesis.[7-10] For instance, the salt dissociation degree can be evaluated as the interaction energy between the lithium cation and the anion, the ion-pair dissociation energy, which strongly depends on the charge distribution in the anion.[11] To achieve the lowest energy of dissociation by the most extensive delocalization of the negative charge within the anion, strongly electron withdrawing groups are applied to the anion structures; e.g. -F, $-CF_3$, -CN, >CO, and $>SO_2$.[12]

Also the ESWs can be evaluated[10,13] and while the oxidation potential limit of the anion unquestionable should be as high as possible to accommodate high voltage cathode active materials, there is no similar strict demand for the reduction potential limit. Even the most used solvents, the organic carbonates, undergo reduction processes at potentials of ca. 1 V *vs.* Li^+/Li°. Therefore, it is essential to provide a kinetic stability against further solvent reduction by forming a stable solid electrolyte interphase (SEI) layer on the anode surface. A salt able to decompose in a controlled manner and create a SEI layer is therefore highly desirable[14] – exemplified by lithium salts with anions based on boron central atoms such as lithium bis(malonato)borate (LiBMB) and lithium bis(oxalato)borate (LiBOB).[15] Also the lithium bis(trifluoromethanesulfonylimide) (LiTFSI) salt forms adequate SEI layers, but only at very high concentrations.[16] Moreover, even the thermal stability can be assessed, including the prediction of decomposition products,

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Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 10.1039/x0xx00000x

and has provided clear explanations for the thermal decomposition mechanism of $LiPF_6$-based electrolytes.[12] Recently, COSMO[18] approaches have been used for electrolyte studies to predict properties such as viscosities, solubilities and melting points[19]

Previously, many variations of the imide $(N(SO_2C_nF_{2n+1})_2^-)$[12], including e.g. the TFSI and the bis(fluorosulfonyl)imide (FSI) anions, and the Hückel $(C_nN_{5-n}(CF_3)_m(CN)_{n-m}^-)$[20,21], including e.g. the 4,5-dicyano-1,2,3-triazolate (DCTA) and the 4,5-dicyano-2-(trifluoromethyl) imidazolide (TDI) anions, families of anions have been in silico screened as PF_6^- alternatives. So far, some experiments suggest that $LiPF_6$ indeed could be replaced by one of these three main concepts, as represented by for example LiTFSI[22], LiBOB[23] or LiTDI[24]. They all exhibit higher thermal stabilities, are resistant to hydrolysis, and do provide sufficient conductivities of their electrolytes. Application of cyano rather than fluorinated ligands, as for TDI, also makes for less environmentally demanding salts – as we avoid fluorine synthesis. Each of these salts, however, has some fundamental drawbacks and common to all is a higher cost of production. Furthermore, LiTFSI does not properly passivate the aluminum current collector, causing its corrosion and degradation during the entire life of a LIB,[25] while LiBOB shows a low solubility in organic carbonate solvents, a critical parameter especially at low operation temperatures.[26] LiTDI in turn, has problems in creating a stable SEI.[27]

Herein, we apply in silico screening to a new anion design concept, combining two current leading ideas: a tetrahedral boron atom as the anion core[28,29,30,31] and a conjugated system as the anion ligands;[24,32] here composed of parts of imidazole and pyrrole rings of Hückel anions. All resulting anions are extensively compared with a wide range of other relevant anions: i) tetrahedral boron-centered anions: BF_4^-, $B(CH_3)_4^-$, BPh_4^- , BISON $[B(CN)_4^-]$, BPFPB $[(C_2(CF_3)_4O_2)_2B^-]$,[23] BBB $[(1,2\text{-}C_6H_4O_2)_2B^-]$,[33] 4F-BBB $[(1,2\text{-}C_6F_4O_2)_2B^-]$,[34] BCB $[(C_5O_5)_2B^-]$,[35] BNB $[(2,3\text{-}C_{10}H_6O_2)_2B^-]$,[34] BBPB $[(2,2'\text{-}C_{12}H_6O_2)_2B^-]$,[36] BSB $[(C_7H_4O_3)_2B^-]$,[37] BOB[38] and BMB[23] ii) conjugated anions: DCTA,[39] TIM $(C_6N_5^-)$,[40] TCP $(C_8N_5^-)$,[40] PCPI $[C_3(CN)_5^-]$,[40] TDI,[41] TDBI $(C_9N_4H_2CF_3^-)$[42] and TDPI $(C_7N_6CF_3^-)$[43] and iii) standard LIB electrolyte anions: PF_6^-, AsF_6^-, and TFSI. All anion data are compared by three different measures: the cation-anion interaction strength, the stability vs. oxidation, and the stability vs. reduction.

2. Computational

Five ligands with conjugated backbones were applied: (1,2-dicyanoethane-1,2-diylidene)diazanide $(C_4N_4^{2-})$, (1,1,1,4,4,4-hexaflurobutane-2,3-diylidene)diazanide $(C_4F_6N_2^{2-})$, (1,2-difluoroethane-1,2-diylidene)diazanide $(C_2F_2N_2^{2-})$, [(Z)-1,2-dicyanoethene-1,2-diyl]diazanidecarbonitrile $(C_6N_6^{2-})$, (1Z,3Z)-buta-1,3-diene-1,2,3,4-tetracarbonitrile $(C_8N_4^{2-})$. For each choice of ligand, three boron-centered anions were built by: i) two ligands of the same type, ii) one ligand and two cyano groups, and iii) one ligand and two fluorine atoms, and hereby in total 15 new anions were generated (Figure 1).

The electronic dissociation energy (ΔE_d) moving from a Li^+-anion ion-pair to separated ions was calculated for all anions using an approach outlined before.[44,45] Initially the geometry of each anion was optimized and subsequently a lithium cation was manually inserted at a wide range of possible coordination sites including mono-, bi- and tri-dentate coordination, creating up to 15 different ion-pairs for each anion, and the geometries of all the resulting ion-pairs were optimized, all using the B3LYP functional[46] and the 6-311+G(d) basis set. The basis set superposition error (BSSE) was neglected in the ion-pair calculations as it was previously found to typically be only a few $kJ \cdot mol^{-1}$.[44] The obtained electron densities were used for an analysis of the charge distribution using the atoms-in-molecules (AIM) theory.[47]

To assess the oxidative electrochemical stability limit (E_{ox}) the vertical ionization potentials (IP) (ΔE_v) were calculated as the differences between the anion and the neutral radical electronic energies, without any geometry relaxation of the radical i.e. change from the anion geometry (the Franck-Condon approximation). For a proper assessment of the

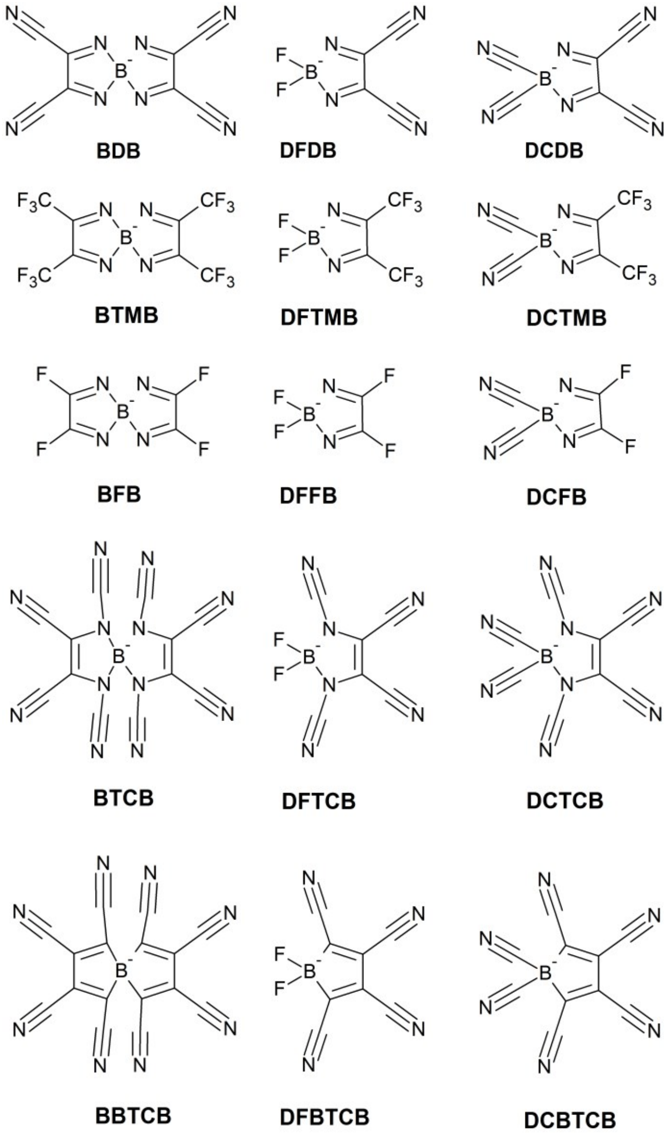

Figure 1. The chemical structures of the proposed anions.

DOI: 10.1039/C6CP02409B

reduction limit (E_{red}) via the electron affinity (EA) it proved necessary to apply a more accurate approach, where the reduced species was relaxed and also the solvent influence was considered by applying the C-PCM SCRF solvation model with all solvent parameters for water, foremost ε = 78. Water is chosen as a simple proxy for a large dielectric solvent. Due to the possibility of ion-pairs being reduced rather than the anions, the reduction potentials were also calculated for the most stable contact ion-pairs. All potentials were adjusted towards the $Li^+/Li°$ reference by -1.46 V.[48] For all electrochemical stability limit calculations the M06-2X functional[49] was used, as it was previously found to provide the best predictive power.[50] This functional was also used to calculate the chemical hardness (η) to evaluate the reactivity of the new species. As defined by Pearson[51], the chemical hardness is the average of the vertical IP and the vertical EA: η = (IP – EA)/2. All computations were made using the Gaussian09 program.[53]

3. Results and discussion

3.1. Cation-anion interactions: Strength and nature

For the various anions our computational scheme arguably neglects both any BSSE and solvent effect influence, in order to enable the best possible comparison to literature data. The main purpose is, however, to create a relative scale/ranking of the anions by creating ion-pairs and this results in a large set of stable structures and in Figure 2 the obtained ΔE_d for the most stable cation-anion ion-pairs and the ΔE_v for the anions, respectively, are collected in a so-called Scheers-plot.[46] The various novel ligands applied to create the anions are easily identifiable by the use of different colors. The use of various -CN groups, in red, blue and purple, rather than -F (green) or -CF_3 (yellow) as ligands, clearly promotes a reduction of the cation-anion interaction energy. All ΔE_d values are collected in

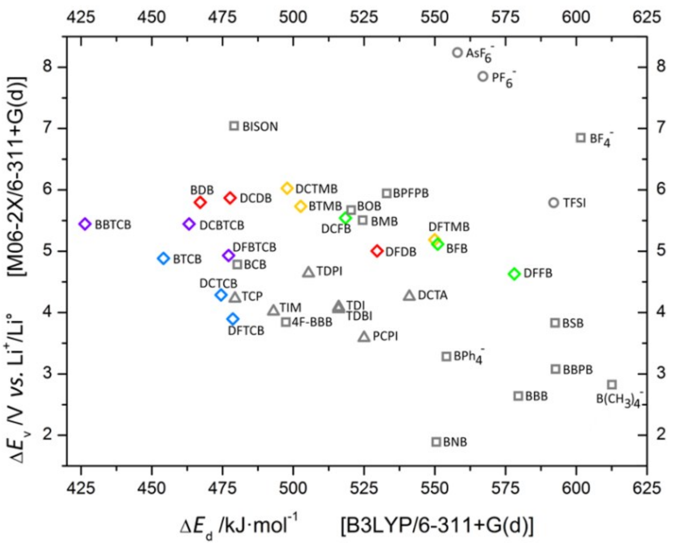

Figure 2. A Scheers-plot showing the ion-pair dissociation energies (ΔE_d) and the anion oxidation potentials (ΔE_v) for the novel anions (diamonds) and a few reference systems; boron centered – squares, Hückel – triangles and standard anions - circles.

Table 1. Application of fluorinated ligands allows ΔE_d values down to 518 kJ·mol^{-1} for DCFB and 498 kJ·mol^{-1} for DCTMB, while using the same ligand backbone but with -CN groups, lowers the ΔE_d significantly, down to 467 kJ·mol^{-1} for BDB. Overall, the conjugated -CN based ligands create weaker cation-anion interactions than the use of conventional -CN groups, and all fluorinated ligands are even more inferior. This can clearly be seen by comparing with ΔE_d (479 kJ·mol^{-1}) for BISON, containing only conventional -CN group ligands.

The introduction of further modifications, such as adding -CN groups to nitrogen atoms to create [(Z)-1,2-dicyanoethene-1,2-diyl]diazanidecarbonitrile ligands or even higher substitution degree by replacing these nitrogen atoms by carbon atoms, results in even lower ΔE_d: 454 and 426 kJ·mol^{-1}, respectively for BTCB and BBTCB. The very low cation-anion interaction strength obtained for the latter anion is in fact lower than for any other LIB electrolyte relevant ion-pair that we know of. The comparable lithium salts from the literature with anions based on a boron central atom, except said BISON, reaches ΔE_d:s of 520-525 kJ·mol^{-1} for BOB and BMB, thus similar but higher, but for many others ΔE_d is closer to 600 kJ·mol^{-1} such as for BF_4^- and $B(CH_3)_4^-$. Also the Hückel salts are not able to compete; their ΔE_d:s range from 479 kJ·mol^{-1} for the pyrolates e.g. TCP, via 493-516 kJ·mol^{-1} for imidazolates e.g. TIM and TDI, to 541 kJ·mol^{-1} for triazolates e.g. DCTA. As for a comparison with the standard PF_6^- anion and its ion-pair with Li^+, ΔE_d is 567 kJ·mol^{-1}.

As the lithium ion is quite small, and thus rendering steric hindrance less likely, the nature of the significant decrease in the interaction energy can primarily be addressed by the anion charge distribution. One crucial parameter, the ligand electron withdrawing force, here measured by the resulting partial charge on the central boron atom, reveals the atom connecting to be the more important design feature, the effect largely stops after the first atom. All resulting charge distributions for the novel anions are listed in Table 2. A fluorine ligand, a very strong electron withdrawing moiety, can cause an increase in the boron atom charge from +1.57 for BH_4^- to +2.36 for BF_4^-, while oxygen connecting ligands result in ca. +2.30, as seen for BOB and BMB, and more medium effects are observed for carbon connecting ligands such as for $B(CH_3)_4^-$ resulting in a boron atom charge of +1.83. However, there are large differences between normal carbon connecting ligands such as -CH_3 and -Ph (ca. +1.8) and -CN ligands (ca. +2.0), probably due to the very strong carbon-nitrogen triple-bond.

The novel anions with a ligand coordination of the boron atom by four nitrogen atoms, as for BDB, BTMB, and BFB, provides an average electron withdrawing force resulting in a boron atom charge of ca. +2.12 (Table 2). This level appears to be the most appropriate, ensuring a fair charge distribution between the core and the ligands, balancing the need for proper delocalization and the risk of creating negatively charged parts/fragments attractive for lithium ion interaction. The change into BTCB by adding -CN groups connected to nitrogen atoms, only slightly increases the observed effect (+2.18). Further replacement of these nitrogen atoms by carbon atoms,

Table 1. The ion-pair dissociation energies, anion oxidation potentials, anion chemical hardness and reduction potentials for anion and ion-pair, of the novel anions/lithium salts and references.

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anion	ΔE_d [kJ·mol^{-1}] B3LYP	ΔE_v [V *vs.* Li$^+$/Li$^\circ$] M06-2X	η [eV] M06-2X	$E_{red\ anion}$ [V *vs.* Li$^+$/Li$^\circ$] M06-2X	$E_{red\ ion\text{-}pair}$ [V *vs.* Li$^+$/Li$^\circ$] M06-2X
BDB	467	5.80	4.48	2.00	1.84
DFDB	530	5.00	4.58	1.85	2.17
DCDB	478	5.87	4.61	2.02	2.36
BTMB	503	5.73	4.97	1.44	1.99
DFTMB	550	5.19	5.11	1.40	1.72
DCTMB	498	6.03	5.09	1.59	1.98
BFB	551	5.12	5.34	0.66	1.01
DFFB	578	4.63	5.50	0.59	0.98
DCFB	518	5.54	5.48	0.83	0.87
BTCB	454	4.88	3.66	1.56	1.67
DFTCB	479	3.90	3.76	1.63	1.75
DCTCB	474	4.29	3.66	1.87	1.92
BBTCB	426	5.44	3.75	2.67	2.82
DFBTCB	477	4.93	3.79	2.48	2.62
DCBTCB	463	5.44	3.75	2.62	2.79
BF_4^-	602	6.85	6.94	-1.32	-0.84
$B(CH_3)_4^-$	613	2.83	4.27	-1.03	-0.74
BPh_4^-	554	3.28	4.35	-	-
BISON	479	7.04	6.29	-0.13	0.23
BPFPB	533	5.94	5.48	-	-
BBB	580	2.64	3.81	-0.64	-0.46
4F-BBB	497	3.84	4.84	-0.57	0.51
BCB	480	4.78	3.80	1.74	2.35
BNB	551	1.89	3.97	0.05	0.27
BBPB	593	3.08	3.95	-0.29	0.03
BSB	593	3.83	4.26	0.13	0.73
BOB	520	5.67	5.37	1.28	1.72
BMB	525	5.51	5.35	-0.74	0.50
DCTA	541	4.26	4.99	0.34	0.51
TIM	493	4.02	4.65	0.41	0.44
TCP	479	4.23	4.33	0.81	0.97
PCPI	525	3.59	3.54	1.89	2.18
TDI	516	4.05	4.74	0.35	0.59
TDBI	516	4.09	4.37	0.64	0.82
TDPI	505	4.64	4.21	1.53	1.74
PF_6^-	567	7.85	7.36	-1.36	-0.57
AsF_6^-	558	8.24	7.16	1.32	1.67
TFSI	592	5.79	5.15	1.22	1.73

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Table 2. The charge distributions for the cores and the ligands of the novel anions.

Anion	Core	Ligands					
	B	-CN	-F	Conjugated			
				Backbone	-CN	-F	$-CF_3$
BDB	+2.12	-	-	-1.03	-0.26	-	-
DFDB	+2.25	-	-0.82	-1.02	-0.29	-	-
DCDB	+2.06	-0.75	-	-1.03	-0.26	-	-
BTMB	+2.12	-	-	-1.15	-	-	-0.21
DFTMB	+2.25	-	-0.82	-1.14	-	-	-0.23
DCTMB	+2.06	-0.76	-	-1.16	-	-	-0.20
BFB	+2.13	-	-	-0.31	-	-0.63	-
DFFB	+2.25	-	-0.83	-0.32	-	-0.63	-
DCFB	+2.07	-0.77	-	-0.27	-	-0.63	-
BTCB	+2.18	-	-	-1.77	+0.28[a] -0.19[b]	-	-
DFTCB	+2.29	-	-0.82	-1.67	+0.25[a] -0.23[b]	-	-
DCTCB	+2.10	-0.76	-	-1.70	+0.27[a] -0.21[b]	-	-
BBTCB	+1.84	-	-	-0.46	-0.25[a] -0.22[b]	-	-
DFBTCB	+2.14	-	-0.81	-0.40	-0.29[a] -0.27[b]	-	-
DCBTCB	+1.92	-0.76	-	-0.42	-0.25[a] -0.24[b]	-	-

[a] nitrile groups close to the boron atom
[b] nitrile groups far from the boron atom

as for BBTCB, results in a decrease to a typical value for carbon connected ligands, +1.84.

As stated above, the part of the ligand not connecting to the central boron atom seems to have no significant direct influence on the core charge. Thus we as a first approximation consider the core and ligand charge distributions independently. As the observed cation-anion interaction energies result from delocalization of the negative charge, comparing BFB, BTMB and BDB, the -F ligand at BFB has the strongest electron withdrawing force with a ligand charge of -0.63, and will thus possibly strongly interact with the lithium cation via this fluorine ligand. In turn, the $-CF_3$ group of BTMB has a much weaker effect, with the charge of trifluoromethyl ligand being equal to -0.21. However, inside the latter ligand there is a significant gradient of charge density: a highly positive carbon atom (+1.61) is surrounded by three strongly electron withdrawing fluorine atoms (-0.61). Thus also the $-CF_3$ ligand can be expected to coordinate strongly to a lithium cation, especially if being able to use two adjacent fluorine atoms in a bi-dentate manner. In contrast, the partial charges of the -CN groups in BDB are -0.26 and this provides the lowest ΔE_d, while the large charge density gradient, +0.91 at C and -1.17 at N, seem to be of less importance and at the same time the lithium cation coordination is mono-dentate.

3.2. Stability against oxidation

The stability *vs.* oxidation is a most important property of anions and boron based anions like BF_4^- and BISON are able to reach 6.85-7.04 V *vs.* $Li^+/Li°$ (Figure 2 and Table 1). However, application of organic ligands, instead of inorganic -F and -CN, causes a drop of the oxidation stability; the best out of these, the BPFPB anion, reaches merely 5.94 V *vs.* $Li^+/Li°$. Notably introduction of cyano groups in general increases the stability of the anion.[12] Here we find that both connecting the -CN moiety directly to the boron core, 6.03 and 5.54 V *vs.* $Li^+/Li°$ for DCTMB and DCFB, respectively, as well as to the backbone, 5.80 and 5.00 V *vs.* $Li^+/Li°$ for BDB and DFDB, respectively, the increase in oxidation potential is observed. Only the addition of additional -CN groups to nitrogen atom causes decrease in ΔE_v: 4.88 V *vs.* Li+/Li° for BTCB. The opposite effect was observed for the substitution by fluorine atoms, where ΔE_v drops to 5.12 V vs. $Li^+/Li°$ for BFB and 4.63 V *vs.* $Li^+/Li°$ for DFFB, respectively.

The exchange of an aromatic system for a corresponding conjugated backbone connected to the tetrahedral boron atom seems to be advantageous for oxidation resistance. Application of the imidazole-derived ligand from the anions TDI and TIM, anions which exhibit ΔE_v:s of 4.05-4.09 V *vs.* $Li^+/Li°$, to the boron-centered anions causes an increased

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stability; 5.80 V *vs.* $Li^+/Li°$ for the BDB anion. Likewise, changing from a typical Hückel anion to a boron-centered using a pyrrole ring derived ligand results in an enhancement of ΔE_v: 4.23 *vs.* 5.44 V *vs.* $Li^+/Li°$, for TCP and BBTCB, respectively. Lithium salts with such high oxidation stabilities are very desirable as they allow for application of cathode materials with higher voltages, and thus improved energy density of the battery. However it should be noted that experimental values of oxidation potential limit (E_{ox}) for Hückel anions (TCP, TIM) are usually higher than predicted with the ΔE_v approach, above 5 V vs. $Li^+/Li°$.[54] On the other hand, ΔE_v:s of boron center anions (BOB, MOB, $B(CH_3)_4^-$)[38,55,56] overestimate E_{ox}:s, which are usually ca. 1 V lower.[50] Despite of these difficulties in correlation, the calculated ΔE_v for anions like BDB, DCDB, BTMB and DCTMB are high enough; 5.73-6.03 V *vs.* $Li^+/Li°$, to consider their application with the cathode material $LiNi_{0.5}Mn_{1.5}O_4$, requiring stability up to 4.7 V *vs.* $Li^+/Li°$.[53] On the other hand, also the salts with the lowest stabilities *vs.* oxidation; DFTCB and DCTCB with ΔE_v:s of 3.90-4.29 V *vs.* $Li^+/Li°$, might still be able to work with standard cathode materials such as $LiFePO_4$ with a mere need of stability up to 3.45 V *vs.* $Li^+/Li°$.[57]

3.3. Stability against reduction

Another significant property is the stability *vs.* reduction, as it is important for all battery electrolyte components to either be stable or enabling the creation of a passivation layer. Especially, a decomposition of a salt anion above 1 V vs. $Li^+/Li°$ can be used to protect the electrolyte carbonate solvents such as EC, PC and DMC against decomposition. The latter is quite typical for boron-based salts, the best example is LiBOB (E_{red} = 1.28 V *vs.* $Li^+/Li°$) that creates SEIs with very good properties.[58] However, even a slight modification of BOB into the BMB anion causes a significant drop of E_{red} to -0.74 V *vs.* $Li^+/Li°$, rendering the latter much less useful for carbonate solvent decomposition protection. In Figure 3 the E_{red} of both the novel anions and their ion-pairs are presented. In most cases the presence of a lithium cation indeed facilitates the process of electron acceptance, the only exception is the BDB anion where the Li^+ coordination slightly stabilizes the structure. There are also a few anions where the anion and the ion-pairs have approx. equal E_{red}: DCFB and DCTCB. In general, however, the differences in E_{red} between the ion-pairs and the anions are of the order 0.1-0.6 V.

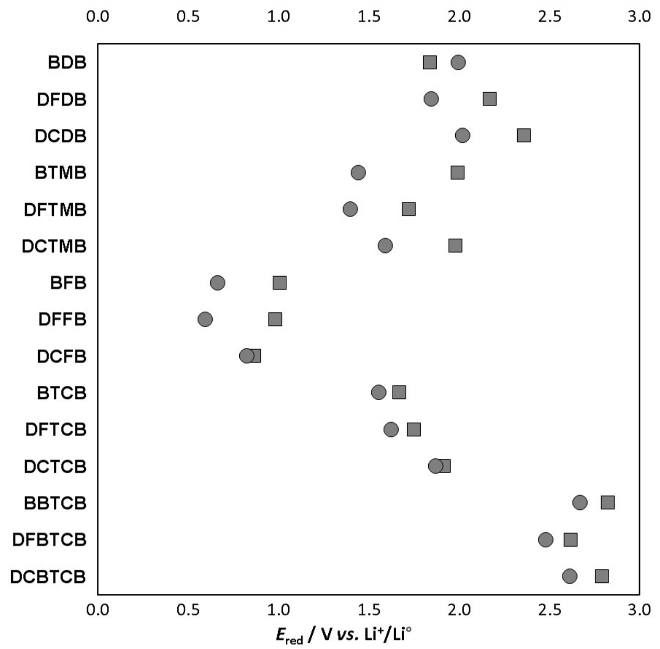

Figure 3. Reduction potentials for anions (circles) and ion-pairs (squares) of the novel lithium salts.

The most stable anions/ion-pairs, i.e. the lowest reduction potentials were obtained for anions containing the (1,2-difluoroethane-1,2-diylidene)diazanide ligand, such as BFB, DFFB and DCFB, 1.01, 0.98 and 0.87 V *vs.* $Li^+/Li°$ for their ion-pairs. Change to conjugated ligands with -CN groups result in larger increases in the reduction ability; 1.84, 2.17, 2.36 V *vs.* $Li^+/Li°$ for the ion-pairs of BDB, BFBD and DCBD, respectively. Application of another -CN group to nitrogen atoms causes a little drop of E_{red}; 1.67 *vs.* $Li^+/Li°$ for BTCB. The least stable anions are those containing the pyrrole-derived ligand – for BBTCB, DFTCB and DCBTCB the reduction potentials all exceed 2.5 V *vs.* $Li^+/Li°$, seemingly preventing any use in LIBs.

3.4. Chemical hardness

An additional parameter that can be explored with a relevance to the requested properties, is the chemical hardness (η) i.e. the resistance of an ion or a molecule to change its charge. It can be used both to explain the level of interaction by donation/acceptance of very small amounts of charge, similar to the HSAB concept,[59] as well as a good approximation of the ability to donate/accept an electron to/from the electrode. The latter means that for anions thought to serve to create an SEI by reduction at the anode, the chemical hardness serves also as a kinetic measure as it is a descriptor of reaction resistance.[60]

For the series of halide anions, F^-, Cl^- and Br^- their η of 6.02, 4.89 and 4.37 eV[50] correlates well with both their ΔE_d:s of 778, 645 and 612 $kJ \cdot mol^{-1}$,[45] and their ΔE_v:s 1.71, 2.23 and 2.03 V *vs.* $Li^+/Li°$.[50] Our results show that the chemical hardness is in a very good agreement with the degree of charge delocalization within the anion (Table 1). The application of fluorine atoms, both as the boron center and as part of conjugated ligand result in higher η; 4.58, 5.34 and 5.50 eV for DFDB, BFB and DFFB, respectively, a little lower values are observed for ligands with $-CF_3$ group; 4.97 and 5.11 eV for BTMB and DDFTMB. The usage of -CN ligands result in lower chemical hardness: down to 4.48 eV for BDB, but the application of up to four -CN groups per ligand results in very low chemical hardness, below 4 eV as shown for BTCB and BBTCB, 3.66 eV and 3.75 eV, respectively. The very low cation-anion interaction energies for the BBTCB and BTCB anions can be explained by this and in general such low levels of chemical hardness are rarely observed, even for completely conjugated and aromatic Hückel anions, only the PCPI pseudo-aromatic anion reaches as low as 3.59 eV. The common LIB anions, such as PF_6^- and TFSI, are above 7 and 5 eV, respectively.

4. Conclusions

In this paper we present an idea of novel anions by combining conjugated ligands and boron centres, to benefit from advantages of both. They all have extensive delocalization of the negative charge and thereby achieve anions with very small lithium-ion interaction energies, even below 430 kJ·mol^{-1}, which is ca. 75% of Li^+-PF_6^-. Moreover, compared to other similar salts e.g. aromatic Hückel anions, the tetrahedral boron centre salts with conjugated ligands should have improved oxidation stabilities, opening for application with high voltage cathodes – even at 5 V vs. Li^+/Li°. The here suggested anions BDB, DCDB and possibly BBTCB, could all provide excellent, easily dissociative salts for high voltage LIB cells. Another very promising anion – BTCB – could be applied in standard LIBs at lower voltages. Most of the new anions are able not only to provide charge carriers for high conductivity of their electrolytes, but also help to stabilize the SEI between the anode and electrolyte by virtue of their tailored reduction voltages, and those who do not, for example DCFB, BFB and DFFB, could possibly rather be applied to anodes such as LTO[61] than the standard graphite. Other important parameters for future battery application, such as solubility and aluminum corrosion protection were not targeted, but nevertheless, even if failing in any of these aspects, the salts can be proposed as SEI forming additives.

Acknowledgements

The authors acknowledge Prof. Michael Armand, CIC Energigune, and the ALISTORE-ERI community for many valuable discussions. All calculations have been carried out at the Wroclaw Centre for Networking and Supercomputing (http://www.wcss.pl), grant No. 346. The support by ALISTORE-ERI for doctoral studies and by Chalmers Area of Advance Energy for travel scholarships to Piotr Jankowski are both gratefully acknowledged. Patrik Johansson acknowledges both the Swedish Energy Agency for a basic research grant via the Swedish Research Council and the continuous support by many of Chalmers Areas of Advance: Energy, Materials Science, and Transport.

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DOI: 10.1039/C6CP02409B

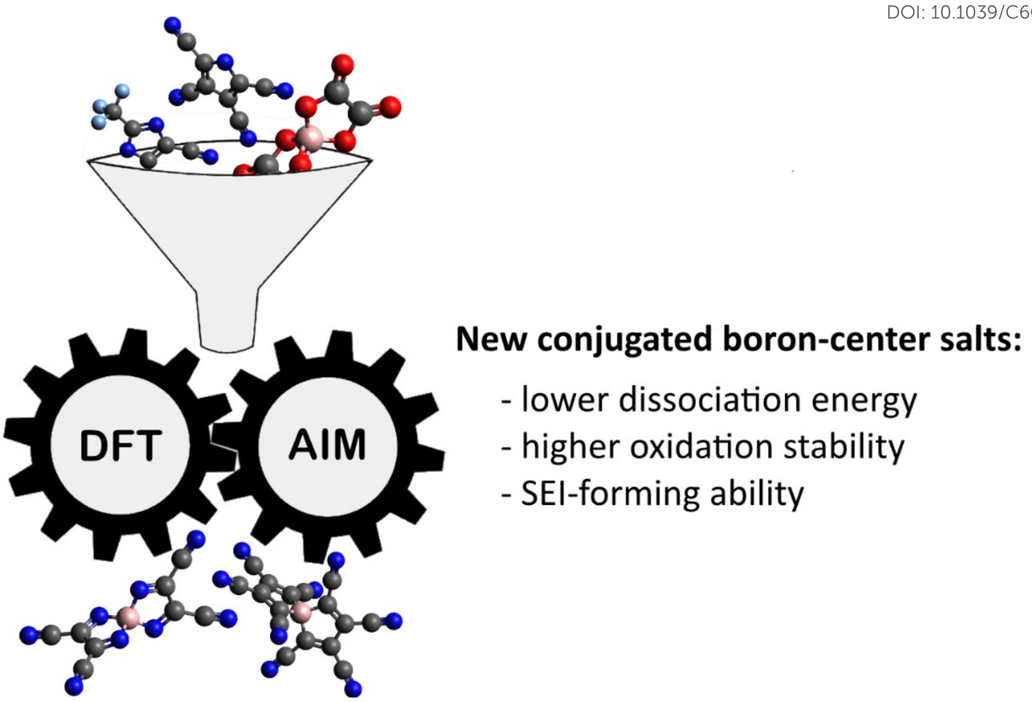

Table of contents: A new lithium salt design concept, based on anions combining a central boron atom and conjugated ligands is presented